

EXPERIMENT – 03

Experiment No.	03
Date	_____
Title	Farm Animals Java Application Using Classes and Objects

1. Objective / Aim

To design a Java application for farm animals such as cow, pig, and horse using classes and objects, and display where they stay, what they eat, and the sounds they make.

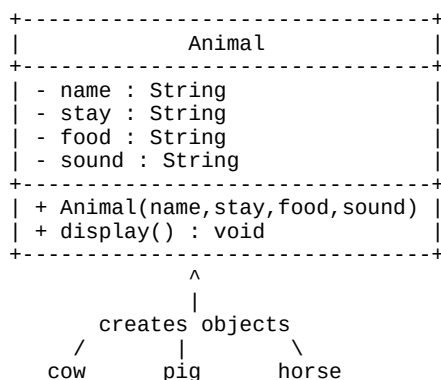
2. System Requirements

Hardware	Computer/Laptop with keyboard and monitor
Software	JDK (Java Development Kit), Command Prompt or any Java IDE

3. Algorithm / Logic Description

1. Start the program.
2. Create an Animal class with name, stay, food, and sound attributes.
3. Create a constructor to initialize the animal details.
4. Define a display() method to print the animal information.
5. In the main() method, create objects for cow, pig, and horse using the Animal class.
6. Call display() for each object.
7. Stop the program.

4. UML Class Diagram



5. Source Code

```
class Animal {
    String name;
    String stay;
    String food;
    String sound;

    Animal(String name, String stay, String food, String sound) {
        this.name = name;
        this.stay = stay;
        this.food = food;
        this.sound = sound;
    }

    void display() {
        System.out.println("Animal : " + name);
        System.out.println("Stays : " + stay);
    }
}
```

```

        System.out.println("Eats    : " + food);
        System.out.println("Sound   : " + sound);
        System.out.println();
    }
}

public class ClassObjLabPro {
    public static void main(String[] args) {

        Animal cow = new Animal("Cow", "Cowshed", "Grass", "Moo");
        Animal pig = new Animal("Pig", "Pigsty", "Vegetables", "Oink");
        Animal horse = new Animal("Horse", "Stable", "Grass", "Neigh");

        cow.display();
        pig.display();
        horse.display();
    }
}

```

6. Compilation & Execution Commands

```

javac ClassObjLabPro.java
java ClassObjLabPro

```

7. Output Screen

Test Case 1 – Farm Animals Details

```

Animal : Cow
Stays  : Cowshed
Eats   : Grass
Sound  : Moo

Animal : Pig
Stays  : Pigsty
Eats   : Vegetables
Sound  : Oink

Animal : Horse
Stays  : Stable
Eats   : Grass
Sound  : Neigh

```

Test Case 2 – Object-wise Verification

```

Cow object    -> Cow, Cowshed, Grass, Moo
Pig object    -> Pig, Pigsty, Vegetables, Oink
Horse object  -> Horse, Stable, Grass, Neigh

```